

Homework 1 Report

The dataset was perturbed using K-anonymity with $k = 3$ and applied using Mondrian Algorithm. The multidimensional algorithm is applied by selecting the quasi identifiers (HomePlanet, CryoSleep, Age, Destination). The algorithm works by selecting the feature with the highest variance to have higher diversity. Then it does partitioning recursively as long as each side has k or more elements. Finally, it does the summarization. Numerical attributes were generalized into intervals, while categorical attributes were generalized into sets of possible values.

The model that is used is the ScikitLearn SVC with the following parameters, `kernel='rbf', C=1.0, gamma='scale'`. Without the k-anonymity the model can generate the following results:

```
The accuracy score is 0.7223903177004538  
The precision score is 0.7525083612040134  
The recall score is 0.672645739910314  
The auc score is 0.722999745912278
```

On the other hand, when k-anonymity is applied the results flunked resulting in the following scores:

```
The accuracy score is 0.546142208774584  
The precision score is 0.5797413793103449  
The recall score is 0.3991097922848665  
The auc score is 0.5490919331794702
```

It can be seen that k-anonymity deeply hurts the model performance since it can reduce the data utility value.